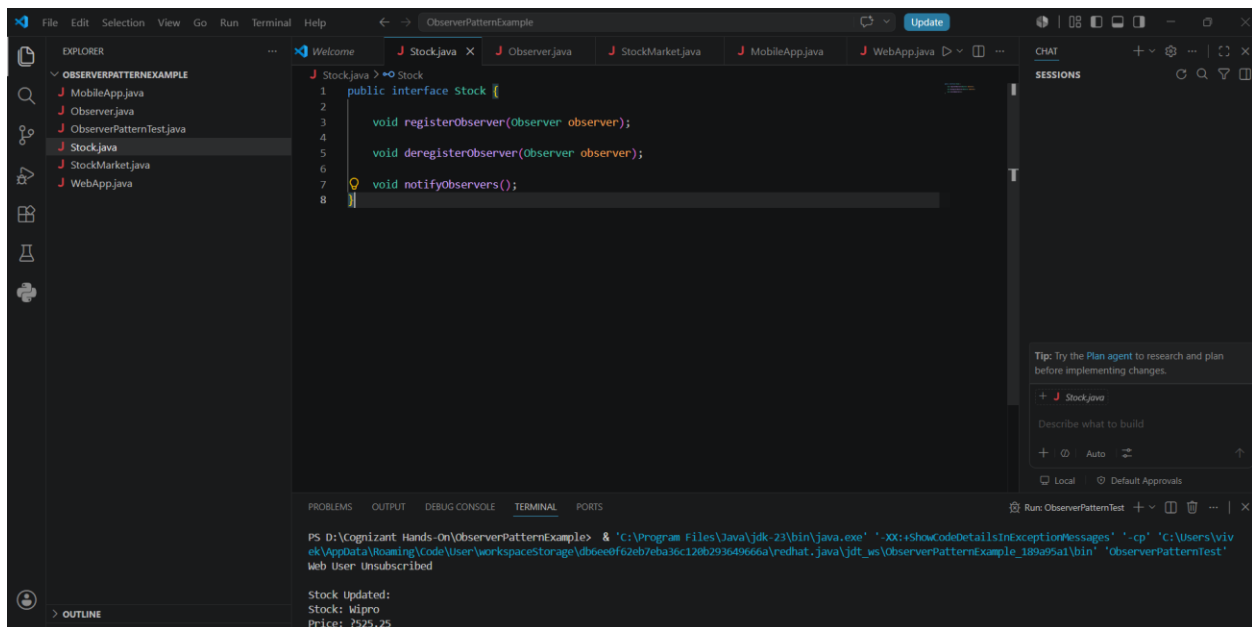


Hands-On

Exercise 7: Implementing the Observer Pattern

As part of this module, I successfully implemented, executed, and tested the Observer Pattern in Java by developing a project named ObserverPatternExample. I created a Stock interface with methods to register, deregister, and notify observers, and implemented a StockMarket class that maintained a list of observers and managed stock price updates. I also defined an Observer interface with an update() method and developed concrete observer classes such as MobileApp and WebApp to receive notifications whenever stock prices changed. After completing the implementation, I executed the program and tested the registration, deregistration, and notification mechanisms to verify that all subscribed observers received updates automatically when stock data was modified. This hands-on exercise helped me understand how the Observer Pattern enables efficient one-to-many communication between objects, promotes loose coupling, and ensures that dependent components remain synchronized with changes in the system.

Coding:



```
File Edit Selection View Go Run Terminal Help
ObserverPatternExample
Update

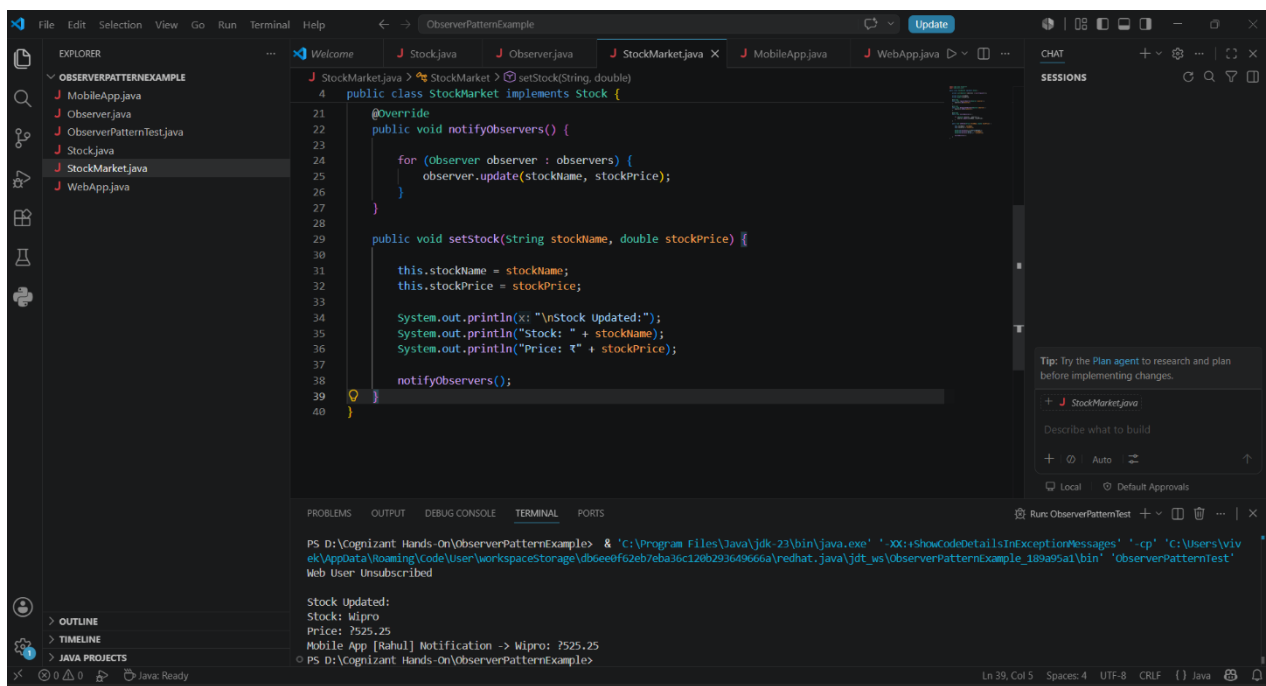
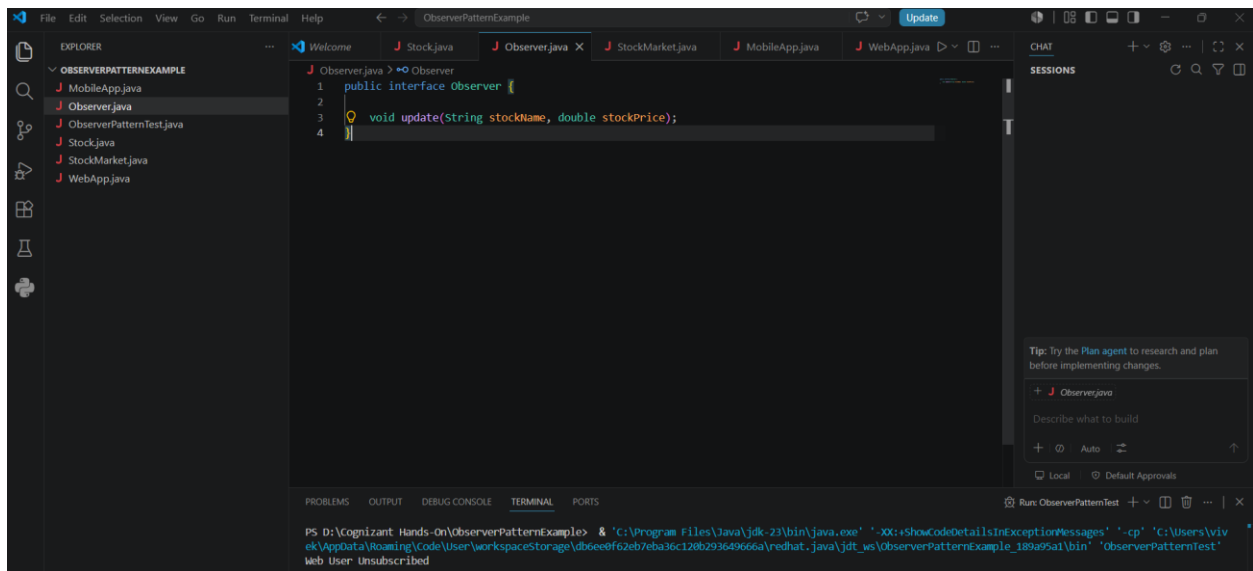
EXPLORER
ObserverPatternExample
  MobileApp.java
  Observer.java
  ObserverPatternTest.java
  Stock.java
  StockMarket.java
  WebApp.java

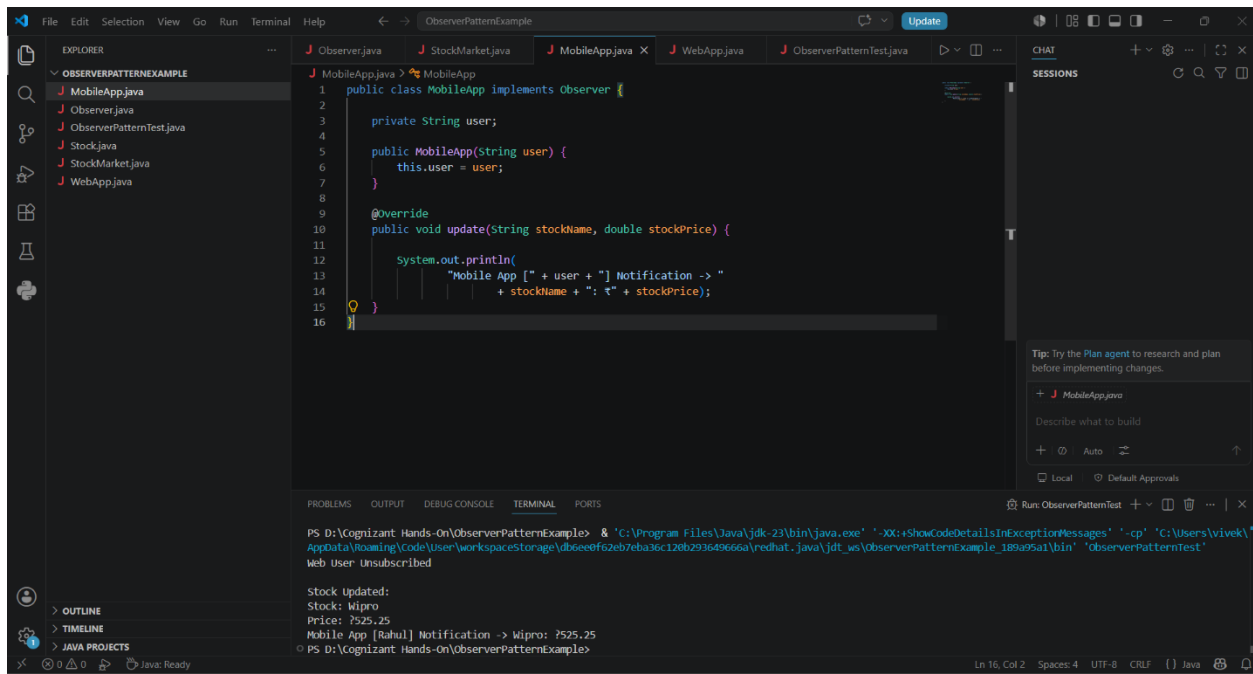
J Stock.java > Stock
1 public interface Stock {
2
3     void registerObserver(Observer observer);
4
5     void deregisterObserver(Observer observer);
6
7     void notifyObservers();
8 }

CHAT
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
+ Stock.java
Describe what to build
+ Auto
Local Default Approvals

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Run: ObserverPatternTest
PS D:\Cognizant Hands-On\ObserverPatternExample> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\viv\ek\AppData\Roaming\Code\User\workspaceStorage\0b6eef62eb7eba36c126b293649666a\redhat.java\jdt_ws\ObserverPatternExample_189a95a1\bin' 'ObserverPatternTest'
Web User Unsubscribed

Stock Updated:
Stock: Wipro
Price: 7525.25
```

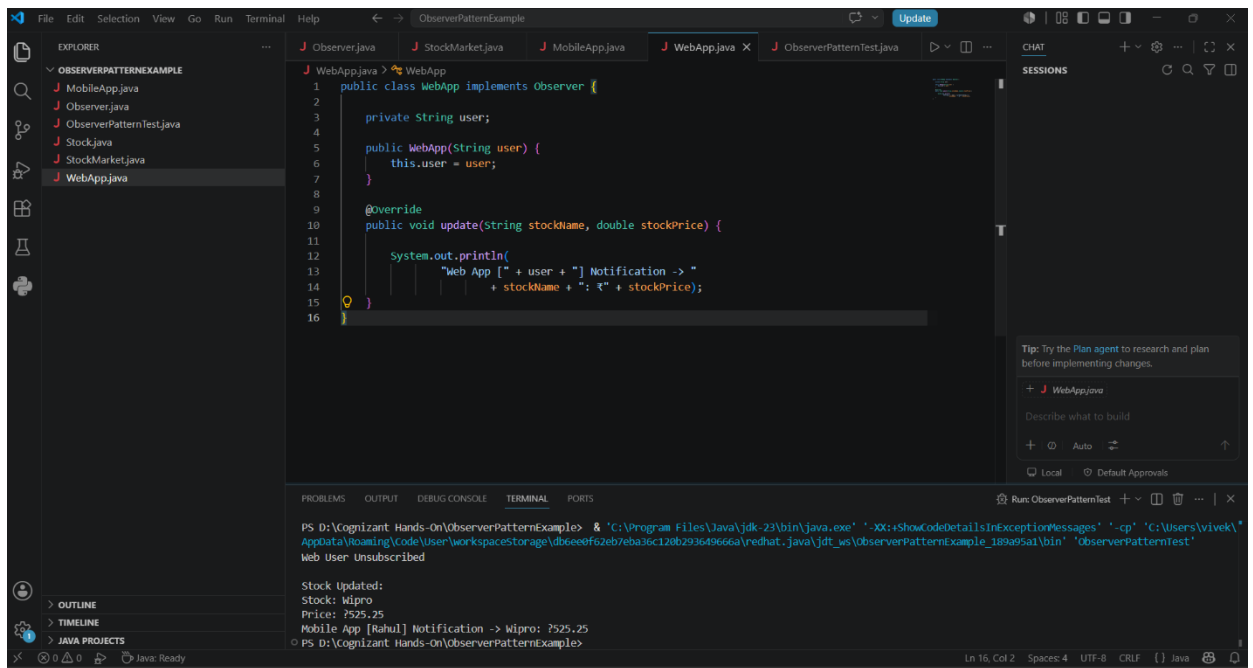




```
File Edit Selection View Go Run Terminal Help
ObserverPatternExample
EXPLORER
ObserverPatternExample
  MobileApp.java
  Observer.java
  ObserverPatternTest.java
  Stock.java
  StockMarket.java
  WebApp.java
MobileApp.java
1 public class MobileApp implements Observer {
2
3     private String user;
4
5     public MobileApp(String user) {
6         this.user = user;
7     }
8
9     @Override
10    public void update(String stockName, double stockPrice) {
11
12        System.out.println(
13            "Mobile App [" + user + "] Notification -> "
14            + stockName + ": ₹" + stockPrice);
15    }
16 }
```

```
PS D:\Cognizant Hands-On\ObserverPatternExample> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\viivek\AppData\Roaming\Code\User\workspaceStorage\db6ee0f62eb7eba36c128b293649666a\redhat.java\jdt_ws\ObserverPatternExample_189a95a1\bin' 'ObserverPatternTest'
Web User Unsubscribed

Stock updated:
Stock: Wipro
Price: 7525.25
Mobile App [Rahul] Notification -> Wipro: 7525.25
PS D:\Cognizant Hands-On\ObserverPatternExample>
```



```
File Edit Selection View Go Run Terminal Help
ObserverPatternExample
EXPLORER
ObserverPatternExample
  MobileApp.java
  Observer.java
  ObserverPatternTest.java
  Stock.java
  StockMarket.java
  WebApp.java
WebApp.java
1 public class WebApp implements Observer {
2
3     private String user;
4
5     public WebApp(String user) {
6         this.user = user;
7     }
8
9     @Override
10    public void update(String stockName, double stockPrice) {
11
12        System.out.println(
13            "Web App [" + user + "] Notification -> "
14            + stockName + ": ₹" + stockPrice);
15    }
16 }
```

```
PS D:\Cognizant Hands-On\ObserverPatternExample> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\viivek\AppData\Roaming\Code\User\workspaceStorage\db6ee0f62eb7eba36c128b293649666a\redhat.java\jdt_ws\ObserverPatternExample_189a95a1\bin' 'ObserverPatternTest'
Web User Unsubscribed

Stock updated:
Stock: Wipro
Price: 7525.25
Mobile App [Rahul] Notification -> Wipro: 7525.25
PS D:\Cognizant Hands-On\ObserverPatternExample>
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Run: ObserverPatternTest + - [ ] ... | X
● PS D:\Cognizant Hands-On\ObserverPatternExample> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\vivek\AppData\Roaming\Code\User\workspaceStorage\db6ee0f62eb7eba36c120b293649666a\redhat.java\jdt_ws\ObserverPatternExample_189a95a1\bin' 'ObserverPatternTest'

Stock Updated:
Stock: TCS
Price: ₹3850.5
Mobile App [Rahul] Notification -> TCS: ₹3850.5
Web App [Priya] Notification -> TCS: ₹3850.5

Stock Updated:
Stock: Infosys
Price: ₹1625.75
Mobile App [Rahul] Notification -> Infosys: ₹1625.75
Web App [Priya] Notification -> Infosys: ₹1625.75

Web User Unsubscribed

Stock Updated:
Stock: Wipro
Price: ₹525.25
Mobile App [Rahul] Notification -> Wipro: ₹525.25
○ PS D:\Cognizant Hands-On\ObserverPatternExample>
```

Ln 23, Col 2 Spaces: 4 UTF-8 CRLF () Java